Unified Biquaternion Theory: A Geometric Unification of General Relativity, Quantum Field Theory, and the Standard Model

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Abstract

The Unified Biquaternion Theory (UBT) is a research framework based on a single biquaternionic field $\Theta(q, \tau)$ over complex time $\tau = t + i\psi$. In the revised local geometric sector the four covariant gradients $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$ form a biquaternionic tetrad. On the real Lorentz slice their central anticommutator defines the metric directly, without a trace, real-part projection, preferred ψ section, or compact-fiber average. This paper records the canonical kinematics and the still-open dynamical bridge: the metric-compatible frame connection is reconstructed from the tetrad and specified torsion, while the UBT action must select the torsion, the exact left/right representation on Θ, Lorentz-slice preservation, curved integrability, and Einstein dynamics. Claims of complete GR or Standard-Model closure are therefore conditional research goals, not axioms.

Lock-in statement: The canonical classical metric is the coefficient of the central identity $\{E_\mu, E_\nu\}_\sharp/2 = g_{\mu\nu}\mathbf{1}$. No projection operation defines the metric. The antisymmetric algebraic product and the commutator of covariant derivatives remain distinct carriers of bivector/spin and curvature information, respectively.

Future-proofing rule: Any future extension of UBT must preserve the central covariant-tetrad identity, distinguish the frame connection from the coordinate connection, and state which results are kinematic, dynamical, conditional, or speculative. No trace, real-part map, phase projection, or compact-fiber average may be reintroduced as the canonical local metric.

Contents

1 Introduction

The unification of General Relativity (GR) and Quantum Field Theory (QFT) remains one of the most significant open problems in theoretical physics. While GR successfully describes gravity as the curvature of spacetime and QFT provides an accurate description of quantum phenomena and the Standard Model (SM) of particle physics, these frameworks are fundamentally incompatible at high energies and small distances.

The Unified Biquaternion Theory (UBT) proposes a radical solution: both spacetime geometry and quantum fields emerge from a more fundamental structure—a biquaternion field $\Theta(q, \tau)$ defined over complex time $\tau = t + i\psi$.

1.1 Historical Context

1.2 Structure of This Article

This article is organized into 12 sections following the canonical structure:

1. Introduction (this section)
2. Biquaternion algebra
3. The Theta field $\Theta(q, \tau)$
4. Complex time $\tau = t + i\psi$
5. Geometry and metric tensor
6. Stress-energy tensor
7. Einstein field equation
8. Quantum Electrodynamics (QED)
9. Quantum Chromodynamics (QCD)
10. Theta-functions and toroidal structure
11. Psychons and consciousness (speculative; see `speculative_extensions/`)
12. Experimental designs and testable predictions

2 Biquaternion Algebra

3 The Theta Field $\Theta(q, \tau)$

4 The Fundamental Biquaternion Field $\Theta(q, \tau)$

4.1 Definition

The fundamental field of Unified Biquaternion Theory is a **biquaternion**:

$$\boxed{\Theta(q, \tau) \in \mathcal{B} = \mathbb{H} \otimes \mathbb{C}} \tag{1}$$

where:

- $q \in \mathcal{B}$ is a biquaternion coordinate (full coordinate model: 8 real parameters)
- $\tau = t + i\psi$ is the canonical complex time
- Biquaternion time $T_B = t + i\psi + j\chi + k\xi$ is a deprecated/historical extension (see `speculative_extensions/`)
- $\mathcal{B} = \mathbb{H} \otimes \mathbb{C}$ denotes the biquaternion algebra
- $\mathbb{H}$ = quaternions with units $\{1, i, j, k\}$

Coordinate bookkeeping note. In this repository, q can denote either (1) the full biquaternionic coordinate $q = z^\mu e_\mu \in \mathbb{C} \otimes \mathbb{H}$, or (2) the real-projected GR coordinate $q_{\mathrm{GR}} = x^\mu e_\mu$ used in the classical GR recovery chain. The canonical complex-time coordinate/fiber is $\tau = z^0 = t + i\psi$. The GR-sector recovery assumes classical complex holomorphy only in tau.

4.1.1 Explicit Biquaternion Form

The field Θ is written explicitly as:

$$\Theta(q, \tau) = \Theta_0(q, \tau) + \Theta_1(q, \tau)i + \Theta_2(q, \tau)j + \Theta_3(q, \tau)k \tag{2}$$

where $\Theta_a(q, \tau) \in \mathbb{C}$ $(a = 0, 1, 2, 3)$ are **complex-valued** component fields.

Important: Θ is a biquaternion (NOT a matrix). Matrix representations are computational tools only.

4.2 Faithful Matrix Representation and Degree Count

A biquaternion has four complex coefficients and therefore eight real components:

$$\Theta = \Theta_0 + \Theta_1\mathbf{i} + \Theta_2\mathbf{j} + \Theta_3\mathbf{k}, \qquad \Theta_a \in \mathbb{C}. \tag{3}$$

A faithful computational representation is a constrained 2×2 complex matrix,

$$\rho(\Theta) = \begin{pmatrix} \Theta_0 + i\Theta_1 & \Theta_2 + i\Theta_3 \\ -\Theta_2 + i\Theta_3 & \Theta_0 - i\Theta_1 \end{pmatrix}. \tag{4}$$

Thus

$$\dim_{\mathbb{C}} \mathbb{B} = 4, \qquad \dim_{\mathbb{R}} \mathbb{B} = 8. \tag{5}$$

A generic 4×4 or 8×8 complex matrix is not a biquaternion. Matrix-valued flavour or Standard-Model extensions, when used, must be denoted separately (for example $\Theta_{\text{ext}} \in \mathrm{Mat}_n(\mathbb{B})$) and must not be silently substituted for the canonical core field.

4.3 Hermitian Involution

For variational and metric formulas we use the full Hermitian involution ‡, which combines complex conjugation with quaternion conjugation and corresponds to matrix Hermitian conjugation in the representation (4):

$$\rho(\Theta^{\ddagger}) = \rho(\Theta)^{\dagger}. \tag{6}$$

The notation must distinguish this operation from quaternion conjugation without complex conjugation.

4.4 Physical Interpretation

The field $\Theta(q, \tau)$ encodes:

1. **Geometric structure**: The covariant tetrad $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$ emerges from Θ. On the classical Lorentz slice its central anticommutator satisfies $\frac{1}{2}(E_\mu^{\sharp} E_\nu + E_\nu^{\sharp} E_\mu) = g_{\mu\nu}\mathbf{1}$, so no trace, real-part, or fiber projection defines the metric
2. **Matter content**: Fermion fields and masses arise from internal structure
3. **Gauge fields**: Electromagnetic and weak/strong interactions emerge from phase structure
4. **Internal auxiliary sector**: Imaginary time component ψ provides dynamics for an internal auxiliary field (speculative interpretations in `speculative_extensions/`)

4.5 Field Equations and Covariant Derivative Status

A compact candidate master equation used in UBT is

$$D^{\dagger} D \Theta(q, \tau) = \kappa \mathcal{T}(q, \tau). \tag{7}$$

Its precise source functional, independent variables, boundary data, and common-action origin remain part of the dynamical closure programme. The geometric first jet entering the metric is

$$E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta. \tag{8}$$

4.5.1 Classical connection versus its action on Θ

Write $E_\mu = i e_\mu{}^0 \mathbf{1} + e_\mu{}^k \mathbf{e}_k$. For every nondegenerate tetrad and specified torsion, the metric-compatible Lorentz-frame connection is kinematically unique,

$$\omega = \mathring{\omega}(e) + K(T). \tag{9}$$

For $T = 0$ this is the Levi–Civita spin connection. The affine connection $\Gamma^\rho{}_{\mu\nu}$ and frame connection $\omega_\mu{}^a{}_b$ are related by tetrad compatibility; neither is a real-part projection of the other.

This reconstruction does not by itself fix how the connection and internal gauge structure act on the biquaternionic field. A naive one-sided regular action is excluded for generic invertible torsion-free curved configurations. The active algebra-native candidate is the two-sided derivative

$$\boxed{D_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu,} \tag{10}$$

with

$$[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}. \tag{11}$$

The exact decomposition of A_μ, B_μ into Lorentz and Standard-Model connections must be derived from the same UBT action. In the invertible torsion-free branch, integrability requires the curvature-intertwiner relation $F^A_{\mu\nu} = \Theta F^B_{\mu\nu}\Theta^{-1}$; this permits nonzero curvature but does not prove existence of a curved solution.

In Cartesian inertial Minkowski gauge one may choose $\Gamma = \omega = \Omega = 0$, hence $D_\mu = \partial_\mu$. The affine field

$$\Theta_{\text{SR}} = \Theta_0 + \sqrt{\mathcal{N}_0}(ix^0\mathbf{1} + x^k\mathbf{e}_k) \tag{12}$$

generates the constant Minkowski tetrad and has vanishing second spacetime derivatives.

The remaining dynamical gaps are torsion selection, the exact paired left/right lift, curved-space existence and uniqueness of $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$, Lorentz-slice preservation by the full dynamics, and derivation of Einstein equations from the canonical action.

4.6 Normalization

The field satisfies the normalization condition:

$$\mathrm{Tr}(\Theta^\dagger\Theta) = \text{const.} \tag{13}$$

for bound states. The constant depends on the physical system under consideration.

4.7 Gauge Transformations

Under local gauge transformations:

$$\Theta(q,\tau) \to U(q,\tau)\Theta(q,\tau)V^\dagger(q,\tau) \tag{14}$$

where $U, V \in U(n)$ are unitary matrices encoding:

- $U(1)$ electromagnetic gauge symmetry
- $SU(2)$ weak isospin symmetry
- $SU(3)$ color symmetry

4.8 Reality Conditions

For physical observables, we impose:

$$\mathrm{Re}\,\mathrm{Tr}(\Theta^\dagger\mathcal{O}\Theta) \in \mathbb{R} \tag{15}$$

for any observable operator $\mathcal{O}$.

4.9 Asymptotic Behavior

At spatial infinity ($|q| \to \infty$):

$$\Theta(q, \tau) \to \Theta_0 + O(1/|q|) \tag{16}$$

where Θ_0 is the vacuum configuration.

For temporal asymptotics (large $|t|$), boundary conditions depend on the specific physical scenario (scattering, bound states, etc.).

4.10 Connection to Biquaternions

The field can be expressed in biquaternion basis:

$$\Theta = \sum_{A=0}^{3} \sum_{B=0}^{3} \theta_{AB}(q, \tau)\, \sigma_A \otimes \sigma_B \tag{17}$$

where σ_A are Pauli matrices (with $\sigma_0 = I$) and θ_{AB} are complex scalar fields.

This representation makes the biquaternionic structure explicit.

4.11 Units and Dimensions

In natural units ($\hbar = c = 1$):

$$[\Theta] = [\text{mass}]^1 = [\text{length}]^{-1} \tag{18}$$

This ensures dimensional consistency with standard field theory.

4.12 Historical Note

This definition supersedes all previous versions found in the repository, including:

- 4D biquaternion representation (old preprint)
- Alternative spinor formulations
- Conflicting matrix dimensions

This is the canonical version. All other formulations should be considered historical or deprecated.

5 Complex Time $\tau = t + i\psi$

6 Biquaternion Time T_B

6.1 Canonical Definition

The fundamental time coordinate in Unified Biquaternion Theory is a **biquaternion**:

$$\boxed{T_B = t + i\psi + j\chi + k\xi} \tag{19}$$

where:

- $t \in \mathbb{R}$ is the **real time coordinate** (standard physical time)
- $\psi, \chi, \xi \in \mathbb{R}$ are **imaginary time components**
- i, j, k are quaternion units satisfying $i^2 = j^2 = k^2 = ijk = -1$

Equivalently, using vector notation:

$$T_B = t + i(\psi + \mathbf{v} \cdot \boldsymbol{\sigma}) = t + i\psi + j\chi + k\xi \tag{20}$$

where $\mathbf{v} = (\chi, \xi, 0)$ and $\boldsymbol{\sigma} = (j, k, ij)$ are quaternion vector components.

Note: The canonical UBT time coordinate is $\tau = t + i\psi$ (complex time). Biquaternion time $T_B = t + i\psi + j\chi + k\xi$ is a deprecated/historical extension preserved here for reference. It is not part of the canonical minimal theory.

6.2 Physical Interpretation

6.2.1 Real Component: Standard Time

The real part t represents:

- Ordinary physical time
- Observable temporal evolution
- Causality structure of spacetime
- Time measured by physical clocks

6.2.2 Scalar Imaginary Component: Isotropic Phase

The scalar imaginary part ψ represents:

- **Isotropic phase structure** of the Θ field
- **Internal auxiliary sector** (speculative; see `speculative_extensions/`)
- **Scalar dark energy** contributions
- **Universal quantum coherence** (direction-independent)

6.2.3 Vector Imaginary Components: Directional Phases

The vector imaginary parts (χ, ξ) represent:

- **Directional phase structures** in spacetime
- **Torsion effects** and spin-orbit coupling
- **Anisotropic dark matter** distributions
- **Directional phase modes** (collective internal sector)

Critical: All components (ψ, χ, ξ) are **dynamical variables** with physical consequences, not mathematical artifacts.

6.3 Dynamics of Imaginary Components

The imaginary time components have their own dynamics. The full equations emerge from variation of the action, but schematically:

$$\Box\psi + m_\psi^2\psi = J_\psi^{\text{scalar}} \tag{21}$$

$$\nabla^2\mathbf{v} + m_v^2\mathbf{v} = \mathbf{J}_v^{\text{vector}} \tag{22}$$

where:

- $\Box = \partial_\mu\partial^\mu$ is the d'Alembertian operator
- m_ψ, m_v are effective mass scales
- $J_\psi, \mathbf{J}_v$ are sources coupling to matter/internal sector
- $\mathbf{v} = (\chi, \xi, 0)$ is the vector imaginary component

These equations couple through the full Θ field dynamics.

6.4 Hierarchical Reduction Structure

6.4.1 Complex Time Limit

When the vector components are negligible ($\|\mathbf{v}\|^2 \ll \psi^2$), or when there is directional isotropy:

$$\boxed{\chi, \xi \to 0 \quad \Rightarrow \quad T_B \to \tau = t + i\psi} \tag{23}$$

This **complex time limit** is valid for:

- Spherically symmetric systems
- Weakly coupled systems
- Weakly-coupled isotropic sector
- Isotropic cosmological backgrounds

6.4.2 Classical Time Limit

When all imaginary components vanish:

$$\psi, \chi, \xi \to 0 \quad \Rightarrow \quad T_B \to t \quad \Rightarrow \quad \text{UBT reduces to standard GR/QFT} \tag{24}$$

This ensures:

- Einstein equations recovered exactly
- Standard Model preserved
- All experimental tests of GR/QFT satisfied

6.4.3 Hierarchy Summary

$$\boxed{T_B = t + i\psi + j\chi + k\xi \quad \xrightarrow{\|\mathbf{v}\|\to 0} \quad \tau = t + i\psi \quad \xrightarrow{\psi \to 0} \quad t} \tag{25}$$

Biquaternion time (full) → **Complex time** (isotropic limit) → **Classical time** (GR limit)

6.5 Measurement and Observability

6.5.1 Direct Observability

The imaginary components (ψ, χ, ξ) are **not directly observable** in classical measurements because:

- Ordinary matter couples only to real metric $g_{\mu\nu} := \mathrm{Re}(\mathcal{G}_{\mu\nu})$ where $\mathcal{G}_{\mu\nu}$ is the biquaternionic metric
- Classical observables involve $\mathrm{Re}\,\mathrm{Tr}(\ldots)$
- Phase information is hidden in quantum coherence

6.5.2 Indirect Detection

The imaginary time components may in principle leave observable traces through:

- Quantum entanglement signatures
- Dark-sector interactions (*speculative/open*)
- Anisotropic dark-sector contributions via vector components (*speculative/open*)
- Torsion effects in strong gravity (vector components)

Consciousness/psychon interpretations are speculative and not part of core UBT. See `speculative_extensions/consciousness/` *for that material.*

6.6 Mathematical Properties

6.6.1 Biquaternion Conjugation

The full conjugate combines complex and quaternion conjugation:

$$T_B^\dagger = t - i\psi - j\chi - k\xi \tag{26}$$

6.6.2 Norm

$$|T_B|^2 = T_B \cdot T_B^\dagger = t^2 + \psi^2 + \chi^2 + \xi^2 \tag{27}$$

6.6.3 Scalar and Vector Parts

$$T_B = t + i\psi_{\text{scalar}} + \mathbf{v}_{\text{vector}} \tag{28}$$

where $\mathbf{v}_{\text{vector}} = j\chi + k\xi$.

6.7 Derivatives and Calculus

6.7.1 Partial Derivatives

For a function $f(T_B) = f(t, \psi, \chi, \xi)$:

$$\frac{\partial f}{\partial T_B} = \frac{1}{4}\left(\frac{\partial f}{\partial t} - i\frac{\partial f}{\partial \psi} - j\frac{\partial f}{\partial \chi} - k\frac{\partial f}{\partial \xi}\right) \tag{29}$$

In the complex time limit ($\chi, \xi \to 0$), this reduces to:

$$\frac{\partial f}{\partial \tau} = \frac{1}{2}\left(\frac{\partial f}{\partial t} - i\frac{\partial f}{\partial \psi}\right) \tag{30}$$

6.7.2 Integration

Biquaternion time integration:

$$\int dT_B = \int dt + i\int d\psi + j\int d\chi + k\int d\xi \tag{31}$$

For physical observables, we typically integrate over real time only:

$$\mathcal{O}_{\text{phys}} = \int dt \, \mathrm{Re}\, \mathcal{O}(T_B)|_{\psi,\chi,\xi=f(t)} \tag{32}$$

6.8 Relation to Other Formulations

6.8.1 Euclidean Time (Wick Rotation)

Biquaternion time T_B is **NOT** the same as Euclidean time from Wick rotation:

- **Wick rotation**: $t \to -it_E$ (analytical continuation)
- **UBT biquaternion time**: $T_B = t + i\psi + j\chi + k\xi$ (physical extension)

In UBT, *all* components t, ψ, χ, ξ are real, physical parameters.

6.8.2 Thermal Field Theory

At finite temperature T, the scalar component ψ may be related to thermal time:

$$\psi \sim \beta = \frac{1}{k_B T} \tag{33}$$

but this is a special case, not a general requirement. The vector components (χ, ξ) are independent of thermal effects.

6.9 Conflict Resolution

This definition supersedes the following conflicting versions:

1. **Drift-diffusion Fokker-Planck variant**: imaginary components as stochastic variables only

2. **Toroidal variant with θ-functions**: time as modular parameter only

3. **Hermitized variant (Appendix F)**: imaginary components as purely mathematical

4. **Complex time only variants**: missing directional structure

Canonical resolution: $T_B = t + i\psi + j\chi + k\xi$ where *all imaginary components* are **dynamical physical fields** with equations of motion. Complex time $\tau = t + i\psi$ is the limiting case valid for isotropic systems.

6.10 Coordinate Systems

6.10.1 Biquaternion Coordinates

The full coordinate set is:

$$(q, T_B) = (q^0, q^1, q^2, q^3, t, \psi, \chi, \xi) \tag{34}$$

where q^μ ($\mu = 0, 1, 2, 3$) are biquaternion spatial coordinates.
In the complex time limit:

$$(q, \tau) = (q^0, q^1, q^2, q^3, t, \psi) \tag{35}$$

6.10.2 Metric Signature

The extended metric in full biquaternion time has signature:

$$\text{signature}(g_{AB}) = (+, -, -, -, -, +, +, +) \quad \text{or similar} \tag{36}$$

depending on convention. The imaginary time coordinates typically have mixed signature from spatial coordinates.

6.11 Conservation Laws

Energy-momentum conservation in biquaternion time:

$$\frac{\partial T^{\mu\nu}}{\partial T_B} = 0 \tag{37}$$

This expands to separate conservation laws for each component:

$$\frac{\partial T^{\mu\nu}}{\partial t} = \text{sources from } \psi, \chi, \xi \tag{38}$$

$$\frac{\partial T^{\mu\nu}}{\partial \psi} = \text{phase sector coupling} \tag{39}$$

$$\frac{\partial T^{\mu\nu}}{\partial \chi}, \frac{\partial T^{\mu\nu}}{\partial \xi} = \text{torsion/anisotropic sources} \tag{40}$$

6.12 Causality Structure

The causal structure is determined by:

$$ds^2 = g_{\mu\nu}(T_B)dx^\mu dx^\nu \tag{41}$$

where $g_{\mu\nu}$ is the central covariant-tetrad metric of the canonical complex-time theory; the T_B construction in this historical extension does not redefine it. Light cones may rotate in the extended time space, but physical causality (in real time t) is preserved through appropriate boundary conditions.

6.13 Symbol Standardization

Symbol	Meaning
T_B	Biquaternion time coordinate (deprecated/historical extension)
τ	Complex time coordinate (isotropic limit)
t	Real time component
ψ	Scalar imaginary time component
χ, ξ	Vector imaginary time components
$T_B^\dagger$	Biquaternion conjugate of T_B

Forbidden uses:

- ψ for wavefunction (use Ψ instead)
- ψ for spinor (use ψ_{spinor} if needed)
- τ for proper time (use s or λ)
- T for temperature (use $\mathcal{T}$ or specify T_{temp})

6.14 Units

In natural units ($\hbar = c = 1$):

$$[t] = [\psi] = [\chi] = [\xi] = [\text{length}] = [\text{time}] \tag{42}$$

All components have dimension of length (or inverse energy).

6.15 Historical Note

The canonical UBT time coordinate is complex time $\tau = t + i\psi$. Biquaternion time $T_B = t + i\psi + j\chi + k\xi$ is a deprecated/historical extension. The imaginary component ψ is a dynamical field; extensions χ, ξ are not part of the canonical minimal theory.

Complex time $\tau = t + i\psi$ is the canonical UBT time coordinate. The earlier biquaternion extensions χ, ξ may appear in exploratory research tracks but are not canonical. This document is preserved for reference; any new canonical content should use τ.

6.16 Simplification: Complex Time Limit

When to use complex time: For systems with directional isotropy or weak coupling, the complex time limit $\tau = t + i\psi$ is sufficient. This includes:

- Spherically symmetric gravitational systems
- Cosmological models with isotropy
- Individual weakly-coupled sector (isotropic modes)
- Weak-field approximations

Biquaternion time T_B (deprecated extension): Extensions χ, ξ have been proposed for strong torsion effects, anisotropic dark matter, and non-Abelian dynamics, but these are speculative and belong in `research_tracks/` or `speculative_extensions/`.

The canonical formulation uses $\tau = t + i\psi$. Biquaternion time T_B is a deprecated extension for research purposes only.

7 Fundamental Biquaternionic Geometry

7.1 Overview

The local classical geometry is carried by

$$E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$$

and the exact central identity

$$\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1}.$$

The symmetric product defines the metric, the antisymmetric algebraic product is a bivector candidate, and the commutator $[D_\mu, D_\nu]$ is the curvature of the frame connection. These three structures must not be conflated.

Canonical geometric hierarchy

1. Fundamental field: $\Theta(q, \tau) \in \mathbb{B}$.
2. Covariant first jet: $E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$.
3. Metric: $\{E_\mu, E_\nu\}_\sharp / 2 = g_{\mu\nu}\mathbf{1}$ on the Lorentz slice.
4. Classical frame connection: for specified torsion, $\omega = \mathring{\omega}(e) + K(T)$ is uniquely determined by the tetrad; the UBT action must select the torsion and its lift to (A_μ, B_μ).
5. Coordinate connection: $\Gamma^\rho{}_{\mu\nu}$, related to Ω_μ by tetrad compatibility.
6. Curvature: $[D_\mu, D_\nu]$ in the relevant representation.

Status: local metric rank, connection reconstruction from specified (e, T), the torsion-free Levi–Civita branch, and flat affine representers are proved. Torsion dynamics, curved integrability, and Einstein dynamics remain open.

Matter-sector caution

The tetrad, metric, connection, and curvature files below implement the current geometric core. The stress-energy and exotic-regime files retain older candidate projection language and are not part of the proved metric theorem.

8 The covariant biquaternionic tetrad E_μ

The fundamental UBT field is $\Theta(q,\tau) \in \mathbb{B}$. The local geometric object is its normalized covariant first derivative

$$\boxed{E_\mu := \mathcal{N}_0^{-1/2} D_\mu \Theta,} \tag{43}$$

where $\mathcal{N}_0 > 0$ is a fixed unit-setting constant. No local denominator is allowed.

8.1 Classical Lorentz sector

Let i be the commuting complex unit and $\mathbf{e}_k$ the quaternion units. The real Lorentz slice is

$$W_L = \{ix^0\mathbf{1} + x^k\mathbf{e}_k \mid x^a \in \mathbb{R}\}. \tag{44}$$

Quaternion conjugation is denoted by $\sharp$ and acts as $(z^0\mathbf{1} + z^k\mathbf{e}_k)^\sharp = z^0\mathbf{1} - z^k\mathbf{e}_k$. The classical tetrad sector is defined by $E_\mu \in W_L$ and $\det(e_\mu{}^a) \neq 0$, where

$$E_\mu = ie_\mu{}^0\mathbf{1} + e_\mu{}^k\mathbf{e}_k. \tag{45}$$

8.2 Metric and bivector from one product

On W_L the symmetrised product is central:

$$\boxed{\frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu) = g_{\mu\nu}\mathbf{1},} \tag{46}$$

with

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1,1,1,1). \tag{47}$$

The metric is therefore not obtained by a trace, real-part projection, phase projection, or compact-fiber average. It is the unique coefficient of the unit element because the full symmetrised product is already central.

The antisymmetric companion

$$\Sigma_{\mu\nu} := \frac{1}{2}(E_\mu^\sharp E_\nu - E_\nu^\sharp E_\mu) \tag{48}$$

is a biquaternionic bivector carrying local oriented-plane/spin information. It is distinct from the connection curvature $[D_\mu, D_\nu]$.

8.3 Transformation and rank

Under coordinate changes E_μ transforms as a covector. Under local Lorentz-frame changes, its coefficients transform by a local Lorentz matrix; Eq. (46) is invariant.

The map $e_\mu{}^a \mapsto g_{\mu\nu}$ has rank ten at every nondegenerate tetrad. Its six-dimensional kernel consists of local Lorentz rotations and boosts. A proof and exact symbolic verification are given in `canonical/gr_closure/covariant_tetrad_rank_theorem.tex` and `tools/verify_covariant_tetrad_rank.py`.

8.4 Status and open bridge

The local metric rank problem is solved at the tetrad level. The metric-compatible frame connection is also kinematically reconstructed from the tetrad and specified torsion, $\omega = \mathring{\omega}(e)+K(T)$; in the torsion-free GR branch this is the unique Levi–Civita spin connection. Every constant Lorentz tetrad has the explicit affine representer $\Theta = \Theta_0 + \sqrt{\mathcal{N}_0}E_\mu x^\mu$ in the flat inertial branch.

For curved configurations, however, it remains to derive torsion and the exact paired left/right connection from the UBT action and to solve the implicit system $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$. A naive one-sided regular action is generically flat for invertible Θ under torsion-free compatibility; the two-sided bimodule derivative avoids this obstruction but does not yet prove curved-space existence.

9 The central anticommutator metric

Let $E_\mu = \mathcal{N}_0^{-1/2}D_\mu\Theta$ be the covariant UBT tetrad. On the classical Lorentz slice W_L, the fundamental quadratic identity is

$$\boxed{E_\mu^\sharp E_\nu = g_{\mu\nu}\mathbf{1} + \Sigma_{\mu\nu},} \tag{49}$$

where

$$g_{\mu\nu}\mathbf{1} = \frac{1}{2}(E_\mu^\sharp E_\nu + E_\nu^\sharp E_\mu), \tag{50}$$

$$\Sigma_{\mu\nu} = \frac{1}{2}(E_\mu^\sharp E_\nu - E_\nu^\sharp E_\mu). \tag{51}$$

The first term is symmetric and central; the second is antisymmetric and biquaternionic. The classical metric is therefore defined intrinsically, not by applying a trace, real-part operator, phase projector, or fiber average to a noncentral object.

In components,

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1,1,1,1). \tag{52}$$

Nondegeneracy is equivalent to invertibility of $e_\mu{}^a$. The local Lorentzian signature follows immediately.

The full product in Eq. (72) remains relevant: its antisymmetric part may carry spin/bivector information. No part is discarded, but no identification with a gauge field is claimed without a transformation and dynamical derivation.

10 The biquaternionic frame connection Ω_μ

A connection specifies how local Lorentz/biquaternionic frames are compared at neighbouring spacetime points. The UBT tetrad is

$$E_\mu = \mathcal{N}_0^{-1/2}D_\mu\Theta. \tag{53}$$

The action of D_μ must always be stated explicitly; left, right, adjoint, and two-sided actions are not interchangeable.

10.1 Classical Lorentz connection reconstructed from tetrad and torsion

Write $E_\mu = ie_\mu{}^0\mathbf{1} + e_\mu{}^k\mathbf{e}_k$. Let

$$T^a = de^a + \omega^a{}_b \wedge e^b = \frac{1}{2}T^a{}_{bc}e^b \wedge e^c. \tag{54}$$

For every nondegenerate tetrad and specified torsion, there is exactly one metric-compatible Lorentz connection,

$$\boxed{\omega^a{}_b = \mathring{\omega}^a{}_b(e) + K^a{}_b(T),} \tag{55}$$

where

$$\mathring{\omega}_\mu{}^a{}_b = e_b{}^\nu \left(\mathring{\Gamma}^\rho{}_{\mu\nu}(g)e_\rho{}^a - \partial_\mu e_\nu{}^a\right) \tag{56}$$

and, with the displayed torsion convention,

$$\boxed{K_{abc} = \frac{1}{2}(T_{cab} - T_{abc} - T_{bca}).} \tag{57}$$

Thus the connection has no residual kinematic ambiguity once (e, T) are fixed. In the torsion-free classical GR branch, $T = 0$, $K = 0$, and $\omega = \mathring{\omega}(e)$.

The biquaternionic/spin connection is the lift

$$\Omega_\mu = \rho_{\mathrm{spin}*}(\omega_\mu) = \frac{1}{4}\omega_{\mu ab}\Sigma^{ab} \tag{58}$$

for the chosen representation. The complete proof is in `canonical/gr_closure/gap_10omega_connecti`

10.2 Relation to Christoffel symbols

The coordinate and frame descriptions are joined by the tetrad postulate,

$$\partial_\mu e_\nu{}^a - \Gamma^\rho{}_{\mu\nu}e_\rho{}^a + \omega_\mu{}^a{}_b e_\nu{}^b = 0. \tag{59}$$

$\Gamma^\rho{}_{\mu\nu}$ transports coordinate indices, while ω and Ω transport local Lorentz/biquaternionic indices. They are related but not identical: they encode the same geometry in different representations. A componentwise identification such as $\Gamma = \operatorname{Re}\Omega$ is generally incorrect.

10.3 Gauge covariance and Lorentz-slice preservation

Under $e \mapsto \Lambda e$,

$$\omega \mapsto \Lambda\omega\Lambda^{-1} - d\Lambda\,\Lambda^{-1}. \tag{60}$$

This is a local Lorentz gauge change, not a new physical degree of freedom. Since every metric-compatible ω_μ lies in $\mathfrak{so}(1,3)$, its parallel transport preserves η_{ab} and the real Lorentz slice, including branches with contorsion. Hence GAP-10L-CONN is closed. GAP-10L-DYN remains open because the complete Θ equations and sources must also preserve the slice.

10.4 Representation selection on Θ

A naive one-sided action

$$D_\mu^L\Theta = \partial_\mu\Theta + A_\mu\Theta \tag{61}$$

obeys $[D_\mu^L, D_\nu^L]\Theta = F_{\mu\nu}^L\Theta$. Under torsion-free tetrad compatibility, invertible Θ would then force $F^L = 0$. Therefore this one-sided regular representation is not the generic curved-GR route.

The algebra-native viable form is the two-sided bimodule derivative

$$\boxed{D_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu,} \tag{62}$$

for which

$$\boxed{[D_\mu, D_\nu]\Theta = F_{\mu\nu}^A\Theta - \Theta F_{\mu\nu}^B.} \tag{63}$$

Torsion-free antisymmetric compatibility requires $F_{\mu\nu}^A\Theta = \Theta F_{\mu\nu}^B$; if Θ is invertible, the two curvatures are conjugate rather than zero. This removes the one-sided flatness obstruction but does not yet derive the precise relation between A_μ and B_μ from the action.

10.5 Flat limit and explicit representer

In a flat inertial torsion-free sector,

$$\Gamma^\rho{}_{\mu\nu} = 0, \qquad \Omega_\mu = 0, \qquad D_\mu = \partial_\mu. \tag{64}$$

The explicit field

$$\Theta_{\mathrm{SR}} = \Theta_0 + \sqrt{\mathcal{N}_0}(ix^0\mathbf{1} + x^k\mathbf{e}_k) \tag{65}$$

generates the Minkowski tetrad and has vanishing second spacetime derivatives.

10.6 Full UBT status

The kinematic connection-reconstruction problem is closed:

- GAP-10Ω-KIN: specified (e, T) uniquely determine ω;
- GAP-10Ω-GR: $T = 0$ gives the Levi–Civita spin connection;
- GAP-10L-CONN: metric-compatible transport preserves the Lorentz slice.

Open dynamical questions are:

- GAP-10T-DYN: derive T from the UBT action and sources;
- fix the exact paired left/right connection and involution on Θ;
- GAP-10I-CURVED: solve the implicit curved system $E_\mu = D_\mu\Theta/\sqrt{\mathcal{N}_0}$;
- GAP-10L-DYN: prove slice preservation by the full dynamics.

11 Biquaternionic curvature

For the viable two-sided biquaternionic derivative

$$D_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu, \tag{66}$$

define

$$F^A_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu], \tag{67}$$
$$F^B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu]. \tag{68}$$

Then

$$\boxed{[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}.} \tag{69}$$

The one-sided formula is recovered only as the special case $B_\mu = 0$.

In the torsion-free compatible branch, antisymmetric integrability requires

$$F^A_{\mu\nu}\Theta = \Theta F^B_{\mu\nu}. \tag{70}$$

For invertible Θ, the curvatures may be nonzero but must be conjugate:

$$F^A_{\mu\nu} = \Theta F^B_{\mu\nu}\Theta^{-1}. \tag{71}$$

This is why the two-sided action avoids the generic flatness obstruction of a naive one-sided regular representation.

The Lorentz frame curvature built from the uniquely reconstructed $\omega = \mathring{\omega}(e) + K(T)$ represents the corresponding Riemann–Cartan geometry. For $T = 0$ it is the usual Riemann curvature of the Levi–Civita connection. The exact identification of (A, B) with the paired spin lifts, and their derivation from the UBT action, remain open.

The flat special-relativistic branch has zero curvature and admits an inertial gauge with $A_\mu = B_\mu = 0$, hence $D_\mu = \partial_\mu$.

Legacy candidate sectors. The files `geometry/biquaternion_stress_energy.tex` and `geometry/exotic_regimes.tex` retain historical candidate projection language. They are excluded from the active compendium until a matter/torsion action consistent with the covariant-tetrad geometry is derived.

12 Classical Geometry (Derived from Biquaternionic Structure)

The following sections present the classical geometric objects through the central covariant-tetrad identity and its compatible connections.

12.1 Classical Metric Tensor

13 The central anticommutator metric

Let $E_\mu = \mathcal{N}_0^{-1/2} D_\mu\Theta$ be the covariant UBT tetrad. On the classical Lorentz slice W_L, the fundamental quadratic identity is

$$\boxed{E^\sharp_\mu E_\nu = g_{\mu\nu}\mathbf{1} + \Sigma_{\mu\nu},} \tag{72}$$

where

$$g_{\mu\nu}\mathbf{1} = \frac{1}{2}(E^\sharp_\mu E_\nu + E^\sharp_\nu E_\mu), \tag{73}$$

$$\Sigma_{\mu\nu} = \frac{1}{2}(E^\sharp_\mu E_\nu - E^\sharp_\nu E_\mu). \tag{74}$$

The first term is symmetric and central; the second is antisymmetric and biquaternionic. The classical metric is therefore defined intrinsically, not by applying a trace, real-part operator, phase projector, or fiber average to a noncentral object.

In components,

$$g_{\mu\nu} = e_\mu{}^a e_\nu{}^b \eta_{ab}, \qquad \eta_{ab} = \mathrm{diag}(-1, 1, 1, 1). \tag{75}$$

Nondegeneracy is equivalent to invertibility of $e_\mu{}^a$. The local Lorentzian signature follows immediately.

The full product in Eq. (72) remains relevant: its antisymmetric part may carry spin/bivector information. No part is discarded, but no identification with a gauge field is claimed without a transformation and dynamical derivation.

13.1 Classical Curvature Tensors

14 Curvature from the connection commutator

The local frame curvature is defined by

$$\mathcal{R}_{\mu\nu}(\Omega) := [D_\mu, D_\nu] = \partial_\mu\Omega_\nu - \partial_\nu\Omega_\mu + [\Omega_\mu, \Omega_\nu]$$

in the selected representation. It measures the failure of a local frame to return unchanged after transport around an infinitesimal loop.

The coordinate Riemann tensor is computed from the compatible affine connection $\Gamma^\rho{}_{\mu\nu}$. The tetrad postulate relates the coordinate and frame curvatures. They are two representations of the same classical local geometry when compatibility and the torsion choice hold; neither is defined as a componentwise real projection of the other.

For the generated tetrad,

$$D_\mu E_\nu - D_\nu E_\mu = [D_\mu, D_\nu]\Theta$$

in the torsion-free coordinate sector. In a flat simply connected region the curvature vanishes and a local frame with $\Omega_\mu = 0$ exists.

Derivation of the Einstein equations from the canonical UBT action remains a separate dynamical gap; this section defines the geometry but does not assume that closure.

14.1 Classical Stress-Energy Tensor

For a specified matter action, the physical stress-energy tensor is defined by Hilbert variation. The older projection-based candidate in `geometry/stress_energy.tex` is not included as a proved canonical result. Conservation and positivity must be established from the complete common action and its equations.

15 Field Equations

15.1 Canonical master equation and status

A commonly used compact form of the UBT matter/master dynamics is

$$D^\dagger D\Theta = \kappa\mathcal{T}. \tag{76}$$

Its precise source functional, connection representation, and common-action origin must be specified before it can be used as a complete gravitational field equation.

The classical metric is already defined kinematically by the central tetrad identity. The remaining dynamical bridge is

$$\text{canonical UBT equations} \quad \Longrightarrow \quad G_{\mu\nu}[g] = \kappa T_{\mu\nu}.$$

This implication is open in the unrestricted canonical theory. In particular, it is not obtained by taking a componentwise real part of a biquaternionic Einstein equation.

Current GR status

Proved locally: the central Lorentz metric and rank-ten tetrad-to-metric map; unique reconstruction of every metric-compatible frame connection from tetrad and specified torsion; the Levi–Civita spin connection in the torsion-free branch; Lorentz-slice preservation by metric-compatible transport; an explicit affine single-Θ representer for every constant Lorentz tetrad, including Minkowski spacetime; the one-sided invertible curved branch is a no-go, while the two-sided derivative has the exact left/right curvature identity.
Open: torsion dynamics from the canonical action; the exact paired left/right connection and involution; local/global existence of the implicit curved system; slice preservation by complete Θ dynamics; and Einstein dynamics from the same action/master equation. Schwarzschild, Kerr, FRW, and gravitational-wave sectors require on-shell canonical construction.

Formal proofs and exact checks are in `gr_closure/gap_10omega_connection_elimination.tex`, `gr_closure/gap_10i_integrability_selection.tex`, `tools/verify_gap_10omega_connection.py` and `tools/verify_gap_10i_integrability.py`.

15.2 Alternative Form with Complex Time

The fundamental equation can also be expressed in terms of the Θ field:

$$\nabla^\dagger\nabla\Theta(q,\tau) = \kappa\mathcal{T}(q,\tau) \tag{77}$$

where:

- the generic curved biquaternionic derivative is two-sided, $D_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu$;
- the Lorentz frame connection is reconstructed as $\omega = \mathring{\omega}(e) + K(T)$, while the canonical action must determine torsion and the paired spin/gauge lifts contained in (A_μ, B_μ);
- Standard Model gauge connections must be embedded consistently in the same left/right action (see `canonical/interactions/sm_gauge.tex`);

- $\tau = t + i\psi$ is complex time (canonical; biquaternion time $T_B = t + i\psi + j\chi + k\xi$ is a deprecated extension)
- $\mathcal{T}(q, \tau)$ is the biquaternionic source term

This form makes explicit the connection to the fundamental Θ field dynamics.

15.3 Cosmological Constant

The cosmological constant can be included:

$$\mathcal{E}_{\mu\nu} + \Lambda \mathcal{G}_{\mu\nu} = \kappa \mathcal{T}_{\mu\nu} \tag{78}$$

In the real limit:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \tag{79}$$

An imaginary-scalar explanation of dark energy is a speculative UBT hypothesis, not a closed consequence of the covariant-tetrad geometry. See Section **??** for details.

15.4 Classical GR Solutions

The canonical program must reproduce the following classical GR sectors on shell; complete derivations are not yet available:

- Schwarzschild metric (spherically symmetric black holes)
- Kerr metric (rotating black holes)
- FLRW metric (cosmology)
- Gravitational wave solutions

16 Quantum Electrodynamics (QED)

17 Quantum Electrodynamics (QED) Lagrangian

17.1 Canonical Definition

The complete QED Lagrangian in the Unified Biquaternion Theory framework is:

$$\boxed{\mathcal{L}_{\text{QED}} = \text{Tr}\left[(D_\mu \Theta)^\dagger (D^\mu \Theta)\right] - \frac{1}{4} F_{\mu\nu} F^{\mu\nu}} \tag{80}$$

where:

- $D_\mu = \partial_\mu + igA_\mu$ is the electromagnetic covariant derivative
- A_μ is the electromagnetic gauge potential (photon field)
- $g = e$ is the electromagnetic coupling constant (elementary charge)
- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor
- $\Theta \in \mathbb{C}^{4\times 4}$ is the biquaternion field

17.2 Components

17.2.1 Matter Field Term

$$\mathcal{L}_{\text{matter}} = \text{Tr}\left[(D_\mu\Theta)^\dagger(D^\mu\Theta)\right] \tag{81}$$

Expanding the covariant derivative:

$$\mathcal{L}_{\text{matter}} = \text{Tr}\left[(\partial_\mu\Theta - igA_\mu\Theta)^\dagger(\partial^\mu\Theta + igA^\mu\Theta)\right] \tag{82}$$

$$= \text{Tr}\left[(\partial_\mu\Theta)^\dagger(\partial^\mu\Theta)\right] + ig\text{Tr}\left[(\partial_\mu\Theta)^\dagger A^\mu\Theta - A_\mu\Theta^\dagger(\partial^\mu\Theta)\right] + g^2A_\mu A^\mu\text{Tr}(\Theta^\dagger\Theta) \tag{83}$$

17.2.2 Photon Field Term

$$\mathcal{L}_{\text{photon}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{84}$$

where the field strength tensor is:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{85}$$

Equivalently, in terms of electric and magnetic fields:

$$\mathcal{L}_{\text{photon}} = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) \tag{86}$$

in natural units with $c = 1$.

17.3 Gauge Symmetry

17.3.1 U(1) Gauge Transformation

The Lagrangian is invariant under local $U(1)$ gauge transformations:

$$\Theta(x) \to e^{i\alpha(x)}\Theta(x) \tag{87}$$

$$A_\mu(x) \to A_\mu(x) - \frac{1}{g}\partial_\mu\alpha(x) \tag{88}$$

where $\alpha(x)$ is an arbitrary real scalar function.

Proof of invariance:

$$D_\mu\Theta \to e^{i\alpha}D_\mu\Theta \quad \Rightarrow \quad (D_\mu\Theta)^\dagger(D^\mu\Theta) \to (D_\mu\Theta)^\dagger(D^\mu\Theta) \tag{89}$$

and $F_{\mu\nu} \to F_{\mu\nu}$ (gauge-invariant).

17.3.2 Gauge Fixing

For quantization, we choose a gauge. Common choices:

- **Lorenz gauge**: $\partial^\mu A_\mu = 0$
- **Coulomb gauge**: $\nabla \cdot \mathbf{A} = 0$
- **Temporal gauge**: $A_0 = 0$

In curved spacetime:

$$\nabla^\mu A_\mu = 0 \quad \text{(covariant Lorenz gauge)} \tag{90}$$

17.4 Equations of Motion

17.4.1 Maxwell Equations

Varying $\mathcal{L}_{\text{QED}}$ with respect to A_μ yields:

$$\partial_\nu F^{\mu\nu} = j^\mu \tag{91}$$

where the electromagnetic current is:

$$j^\mu = ig\text{Tr}\left[\Theta^\dagger D^\mu\Theta - (D^\mu\Theta)^\dagger\Theta\right] \tag{92}$$

In curved spacetime:

$$\nabla_\nu F^{\mu\nu} = j^\mu \tag{93}$$

17.4.2 Bianchi Identity

The field strength satisfies:

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0 \tag{94}$$

Equivalently:

$$\nabla_{[\lambda} F_{\mu\nu]} = 0 \tag{95}$$

17.4.3 Theta Field Equation

Varying with respect to Θ:

$$D^\mu D_\mu\Theta = 0 \tag{96}$$

This is the Klein-Gordon-like equation for the charged biquaternion field.

17.5 Interaction Term

The interaction between matter and electromagnetic field is:

$$\mathcal{L}_{\text{int}} = ig\text{Tr}\left[A^\mu\left(\Theta^\dagger\partial_\mu\Theta - (\partial_\mu\Theta)^\dagger\Theta\right)\right] + g^2 A_\mu A^\mu \text{Tr}(\Theta^\dagger\Theta) \tag{97}$$

The first term is the minimal coupling (current-photon interaction). The second term is the Proca-like mass term if Θ has non-zero vacuum expectation value.

17.6 Current Conservation

The electromagnetic current satisfies:

$$\partial_\mu j^\mu = 0 \tag{98}$$

In curved spacetime:

$$\nabla_\mu j^\mu = 0 \tag{99}$$

This follows from gauge invariance via Noether's theorem.

17.7 Coupling to Complex Time

For complex time $\tau = t + i\psi$, the Lagrangian extends to:

$$\mathcal{L}_{\text{QED}}(\tau) = \text{Tr}\left[(D_\mu\Theta(\tau))^\dagger(D^\mu\Theta(\tau))\right] - \frac{1}{4}F_{\mu\nu}(\tau)F^{\mu\nu}(\tau) \tag{100}$$

where all fields depend on $\tau = t + i\psi$.

The imaginary time component ψ introduces:

- Phase modulation of electromagnetic interactions
- Coupling to internal auxiliary sector (speculative; see `speculative_extensions/`)
- Nonlocal quantum correlations

17.8 Fine Structure Constant

The electromagnetic coupling strength is characterized by:

$$\alpha = \frac{g^2}{4\pi\hbar c} = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137.036} \tag{101}$$

In UBT, α is **predicted** from geometric and topological properties of the Θ field (see Section **??**).

17.9 Renormalization

17.9.1 Running Coupling

The effective coupling α depends on energy scale Q:

$$\alpha(Q^2) = \frac{\alpha(\mu^2)}{1 - \frac{\alpha(\mu^2)}{3\pi}\ln\left(\frac{Q^2}{\mu^2}\right)} \tag{102}$$

where μ is a reference scale.

17.9.2 Charge Renormalization

The bare charge g_0 relates to renormalized charge g via:

$$g_0 = Z_3^{-1/2} g \tag{103}$$

where Z_3 is the photon field renormalization constant.

17.10 Energy-Momentum Tensor

The QED contribution to the stress-energy tensor is:

$$T_{\mu\nu}^{\text{QED}} = T_{\mu\nu}[\Theta] + T_{\mu\nu}^{\text{EM}} \tag{104}$$

where:

$$T_{\mu\nu}^{\text{EM}} = \frac{1}{4\pi}\left(F_{\mu\alpha}F_\nu{}^\alpha - \frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\right) \tag{105}$$

is the electromagnetic stress-energy tensor.

17.11 Curved Spacetime Extension

In curved spacetime with the central covariant-tetrad metric $\{E_\mu, E_\nu\}_\sharp/2 = g_{\mu\nu}\mathbf{1}$:

$$\mathcal{L}_{\text{QED}}^{\text{curved}} = \sqrt{-g}\left[\text{Tr}\left[(D_\mu\Theta)^\dagger(D^\mu\Theta)\right] - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right] \tag{106}$$

where $g = \det(g_{\mu\nu})$ and indices are raised/lowered with $g^{\mu\nu}$ and $g_{\mu\nu}$.
The covariant derivative on Θ includes both gauge and gravitational connections:

$$D_\mu\Theta = \partial_\mu\Theta + igA_\mu\Theta + \Gamma_\mu\Theta \tag{107}$$

where Γ_μ is the spin connection.

17.12 Conflict Resolution

This canonical definition supersedes:

1. **Lagrangian without complex time integration**: Incomplete
2. **E/B from Maxwell in flat space**: Inconsistent with curved spacetime
3. **Alternative gauge choices**: Use Lorenz gauge as default

Canonical resolution: Use Eq. 80 with:

- Covariant derivatives in curved spacetime
- $F_{\mu\nu}$ and $g_{\mu\nu} := \text{Re}(\mathcal{G}_{\mu\nu})$ from biquaternionic metric Eq. **??**
- Complex time τ enters through $\Theta(q, \tau)$

17.13 Computational Implementation

For numerical calculations:

1. Define $\Theta(x, t, \psi) \in \mathbb{C}^{4\times4}$
2. Compute $D_\mu\Theta = \partial_\mu\Theta + igA_\mu\Theta$
3. Evaluate $\text{Tr}[(D_\mu\Theta)^\dagger(D^\mu\Theta)]$
4. Compute $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$
5. Assemble $\mathcal{L}_{\text{QED}}$

17.14 Physical Predictions

From this Lagrangian, UBT **aims to derive** the following (see `canonical/qed_phi_const/` for current status):

- **Electron mass**: $m_e \approx 0.511$ MeV reproduced using fitted parameter $R = 1/m_e$ (**not a zero-parameter prediction**)
- **Fine structure constant**: $\alpha \approx 1/137.036$ (**semi-empirical**; structural vacuum $n^* = 137$ proved, coupling B not yet derived from first principles)
- **Magnetic moment**: Anomalous magnetic moment of electron (Schwinger term $a_e = \alpha/(2\pi)$ **proved** at $\varphi = \text{const}$)
- **Lamb shift**: Energy level corrections in hydrogen (**sketch** — explicit path-integral computation pending, Gap L1)

All standard QED predictions are recovered in the limit $\psi \to 0$.

17.15 Extensions

17.15.1 QED + Weak Interactions

For electroweak unification, extend to:

$$\mathcal{L}_{\text{EW}} = \text{Tr}[(D_\mu\Theta)^\dagger(D^\mu\Theta)] - \frac{1}{4}W^a_{\mu\nu}W^{a\mu\nu} - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{108}$$

where $W^a_{\mu\nu}$ are the weak field strengths and D_μ includes $SU(2)_L$ covariant derivative.

17.15.2 QED + Gravity

Full gravitational coupling:

$$\mathcal{L}_{\text{QED+GR}} = \sqrt{-g}\left[\frac{R}{16\pi G} + \text{Tr}[(D_\mu\Theta)^\dagger(D^\mu\Theta)] - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}\right] \tag{109}$$

where R is the Ricci scalar.

17.16 Historical Note

This canonical QED Lagrangian provides a complete, consistent formulation of electromagnetism within UBT, incorporating complex time, curved spacetime, and biquaternion field structure while maintaining compatibility with standard QED.

18 Quantum Chromodynamics (QCD)

19 Quantum Chromodynamics (QCD) Lagrangian

Status: Embedded — Track A (involution approach) canonical
This section embeds standard QCD in UBT-compatible notation.

$SU(3)$ derivation: $SU(3)_c$ is **Proved** [**L0**] from $\mathbb{C} \otimes \mathbb{H}$ via the involution P_2 and carrier space $V_c = \ker(P_2 + \mathbf{1}) \cong \mathbb{C}^3$ (Theorems G.A–G.D; see `consolidation_project/SU3_derivation/`). An alternative route via octonionic extension ($\mathbb{C} \otimes \mathbb{O}$, Track B) also works but is not required; see `research_tracks/octonionic_completion/`.

Algebraic confinement: Free quarks are **Conjectured** [**L0**] to be algebraically inadmissible in $\mathcal{H}_{\text{phys}} = \text{Im}(\Pi_{\text{color}})$ (colour-singlet condition on the 3-qubit colour space $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$). All colour-neutral hadrons (baryons, mesons, tetraquarks, pentaquarks) satisfy the admissibility condition; isolated quarks do not. See `consolidation_project/confinement/algebraic_confinement.tex`.
Note: this is distinct from the Clay Millennium Prize (Yang–Mills mass gap); UBT does not claim to solve that problem.

19.1 Canonical Definition

Metric convention. The physical metric is defined by the central covariant-tetrad identity $\{E_\mu, E_\nu\}_\sharp/2 = g_{\mu\nu}\mathbf{1}$. Projected-metric formulas in earlier drafts are superseded.

The complete QCD Lagrangian in the Unified Biquaternion Theory framework is:

$$\boxed{\mathcal{L}_{\text{QCD}} = \text{Tr}\left[(D_\mu\Theta)^\dagger(D^\mu\Theta)\right] - \frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu}} \tag{110}$$

where:

- $D_\mu = \partial_\mu + ig_sT^aG^a_\mu$ is the QCD covariant derivative
- G^a_μ is the gluon field for color index $a = 1, \ldots, 8$
- T^a are the $SU(3)$ generators (Gell-Mann matrices)
- g_s is the strong coupling constant
- $G^a_{\mu\nu}$ is the gluon field strength tensor

19.2 Gluon Field Strength

The non-Abelian field strength tensor is:

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc}G^b_\mu G^c_\nu \tag{111}$$

where f^{abc} are the $SU(3)$ structure constants.

The last term represents gluon self-interaction (a key feature of non-Abelian gauge theories).

19.3 SU(3) Color Symmetry

19.3.1 Gauge Group

QCD is based on the gauge group:

$$G_{\text{color}} = SU(3)_c \tag{112}$$

19.3.2 Generators

The $SU(3)$ generators T^a satisfy:

$$[T^a, T^b] = i f^{abc} T^c \tag{113}$$

$$\text{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab} \tag{114}$$

19.3.3 Gell-Mann Matrices

The standard representation uses the eight Gell-Mann matrices λ^a ($a = 1, \ldots, 8$):

$$T^a = \frac{\lambda^a}{2} \tag{115}$$

Explicit forms (for reference):

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{116}$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \tag{117}$$

$$\lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \lambda^8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix} \tag{118}$$

19.4 Color Indices

Standard convention:

- Latin indices $a, b, c = 1, 2, \ldots, 8$ for adjoint representation (gluons)
- Latin indices $i, j, k = 1, 2, 3$ for fundamental representation (quarks)
- Summation convention applies: repeated indices are summed

19.5 Structure Constants

The $SU(3)$ structure constants f^{abc} are totally antisymmetric:

$$f^{abc} = -f^{bac} = -f^{acb} \tag{119}$$

Non-zero values (selection):

$$f^{123} = 1, \quad f^{147} = f^{246} = f^{257} = f^{345} = \frac{1}{2} \tag{120}$$

$$f^{156} = f^{367} = -\frac{1}{2}, \quad f^{458} = f^{678} = \frac{\sqrt{3}}{2} \tag{121}$$

19.6 Gauge Transformations

19.6.1 Local SU(3) Transformation

The Lagrangian is invariant under local $SU(3)$ gauge transformations:

$$\Theta(x) \to U(x)\Theta(x)U^\dagger(x) \tag{122}$$

$$G^a_\mu(x)T^a \to U(x)G^a_\mu(x)T^aU^\dagger(x) - \frac{i}{g_s}(\partial_\mu U(x))U^\dagger(x) \tag{123}$$

where $U(x) \in SU(3)$ is a local gauge transformation:

$$U(x) = \exp\left(i\alpha^a(x)T^a\right) \tag{124}$$

with $\alpha^a(x)$ arbitrary real scalar functions.

19.7 Equations of Motion

19.7.1 Yang-Mills Equations

Varying $\mathcal{L}_{\mathrm{QCD}}$ with respect to G^a_μ yields:

$$D_\nu G^{a\mu\nu} = j^{a\mu} \tag{125}$$

where the color current is:

$$j^{a\mu} = g_s \mathrm{Tr}\left[T^a\left(\Theta^\dagger D^\mu\Theta - (D^\mu\Theta)^\dagger\Theta\right)\right] \tag{126}$$

The covariant derivative of the field strength is:

$$D_\nu G^{a\mu\nu} = \partial_\nu G^{a\mu\nu} + g_s f^{abc} G^b_\nu G^{c\mu\nu} \tag{127}$$

19.7.2 Bianchi Identity

$$D_{[\lambda} G^a_{\mu\nu]} = 0 \tag{128}$$

Explicitly:

$$D_\lambda G^a_{\mu\nu} + D_\mu G^a_{\nu\lambda} + D_\nu G^a_{\lambda\mu} = 0 \tag{129}$$

19.8 Asymptotic Freedom

19.8.1 Running Coupling

The strong coupling $\alpha_s = g_s^2/(4\pi)$ runs with energy scale Q:

$$\alpha_s(Q^2) = \frac{\alpha_s(\mu^2)}{1 + \frac{\beta_0}{4\pi}\alpha_s(\mu^2)\ln(Q^2/\mu^2)} \tag{130}$$

where the one-loop beta function coefficient is:

$$\beta_0 = 11 - \frac{2n_f}{3} \tag{131}$$

with n_f the number of active quark flavors.

For $SU(3)$, $\beta_0 = 11-2n_f/3 > 0$ (assuming $n_f \leq 16$), leading to **asymptotic freedom**:

$$\alpha_s(Q^2) \to 0 \quad \text{as} \quad Q^2 \to \infty \tag{132}$$

19.8.2 QCD Scale Λ_{QCD}

The dimensional transmutation scale:

$$\Lambda_{\text{QCD}} \approx 200 \text{ MeV} \tag{133}$$

In UBT, this is **predicted** from emergent $SU(3)$ structure (see Section 19.15).

19.9 Confinement

At low energies ($Q \lesssim \Lambda_{\text{QCD}}$), the strong coupling becomes large:

$$\alpha_s(Q^2) \to \infty \quad \text{as} \quad Q^2 \to \Lambda_{\text{QCD}}^2 \tag{134}$$

This leads to **color confinement**: quarks and gluons are confined within hadrons.

UBT provides a geometric mechanism for confinement via complex time dynamics (see Section **??**).

19.10 Quark Masses

The QCD Lagrangian includes quark mass terms:

$$\mathcal{L}_{\text{mass}} = -\sum_f m_f \bar{q}_f q_f \tag{135}$$

where f labels quark flavors (up, down, strange, charm, bottom, top) and q_f are quark fields.

In UBT, quark masses emerge from Θ field structure:

- $m_u \approx 2.2$ MeV
- $m_d \approx 4.7$ MeV
- $m_s \approx 95$ MeV
- $m_c \approx 1.28$ GeV

- $m_b \approx 4.18$ GeV
- $m_t \approx 173$ GeV

19.11 Gluon Condensate

The QCD vacuum has non-zero gluon condensate:

$$\langle 0|\frac{\alpha_s}{\pi} G^a_{\mu\nu} G^{a\mu\nu}|0\rangle \approx 0.012 \text{ GeV}^4 \tag{136}$$

This contributes to hadron masses and QCD vacuum energy.

19.12 Chiral Symmetry Breaking

For light quarks ($m_u, m_d, m_s \ll \Lambda_{\text{QCD}}$), the Lagrangian has approximate chiral symmetry:

$$SU(n_f)_L \times SU(n_f)_R \tag{137}$$

which is spontaneously broken to:

$$SU(n_f)_V \tag{138}$$

leading to Goldstone bosons (pions, kaons, eta).

19.13 Energy-Momentum Tensor

The QCD contribution to stress-energy:

$$T^{\text{QCD}}_{\mu\nu} = T_{\mu\nu}[\Theta] + T^{\text{gluon}}_{\mu\nu} \tag{139}$$

where:

$$T^{\text{gluon}}_{\mu\nu} = \frac{1}{4\pi}\left(G^a_{\mu\alpha} G^{a\nu\alpha} - \frac{1}{4} g_{\mu\nu} G^a_{\alpha\beta} G^{a\alpha\beta}\right) \tag{140}$$

19.14 Curved Spacetime Extension

In curved spacetime:

$$\mathcal{L}^{\text{curved}}_{\text{QCD}} = \sqrt{-g}\left[\text{Tr}\left[(D_\mu\Theta)^\dagger (D^\mu\Theta)\right] - \frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu}\right] \tag{141}$$

The covariant derivative includes both color and gravitational connections.

19.15 Emergent SU(3) in UBT

In UBT, the $SU(3)$ color symmetry appears upon octonionic extension ($\mathbb{C} \otimes \mathbb{O}$) of the internal structure of the Θ field:

19.15.1 Mechanism

The 8×8 extended Θ field decomposes into:

$$\Theta_{8\times 8} = \sum_{a=1}^{8} \Theta_a \otimes T^a \tag{142}$$

where Θ_a are 3×3 blocks and T^a are $SU(3)$ generators.

19.15.2 Derivation

The emergent gauge fields G^a_μ arise from phase gradients:

$$G^a_\mu = \frac{1}{g_s}\mathrm{Tr}\left[T^a\left(\Theta^\dagger \partial_\mu \Theta - (\partial_\mu \Theta^\dagger)\Theta\right)\right] \tag{143}$$

This produces $SU(3)$ gauge structure via octonionic extension (Track B hypothesis).

19.16 Conflict Resolution

This canonical definition supersedes:

1. **Appendix G (emergent SU(3))**: Inconsistent color indices
2. **Appendix K5 ($\Lambda_{\mathbf{QCD}}$)**: Different normalization
3. **Old main article text**: Incompatible with complex time

Canonical resolution: Use Eq. 110 with:

- Standard $SU(3)$ generators (Gell-Mann matrices)
- Color indices $a, b, c = 1, \ldots, 8$
- Normalization $\mathrm{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$
- Emergent mechanism from Θ field structure

19.17 Experimental Predictions

From this Lagrangian, UBT provides a **structural framework** for (derivations remain open):

- $\Lambda_{\mathrm{QCD}} \approx 200$ MeV — structural interpretation, not yet derived from $S[\Theta]$
- Quark masses — **open problem**; no first-principles derivation
- **Confinement scale**: Related to ψ dynamics (structural conjecture)
- **Glueball spectrum**: From pure glue sector (open problem)

19.18 Lattice QCD Comparison

Predictions can be tested against lattice QCD simulations:

- Hadron masses
- Glueball masses
- String tension
- Phase transitions

19.19 Historical Note

This canonical QCD Lagrangian provides the strong interaction theory within UBT, where $SU(3)$ color symmetry appears via octonionic extension $\mathbb{C} \otimes \mathbb{O}$ (Octonionic Completion Hypothesis, Track B) rather than being postulated. Derivation from the associative sector $\mathbb{C} \otimes \mathbb{H}$ alone is an open research question (Track A).

20 Standard Model Gauge Structure

21 Standard Model Gauge Structure

21.1 Canonical Definition

The complete Standard Model gauge Lagrangian in the Unified Biquaternion Theory framework is:

$$\boxed{\mathcal{L}_{\text{SM}} = \text{Tr}\left[(D_\mu\Theta)^\dagger(D^\mu\Theta)\right] - \frac{1}{4}W^i_{\mu\nu}W^{i\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} - \frac{1}{4}G^a_{\mu\nu}G^{a\mu\nu}} \tag{144}$$

where the covariant derivative is:

$$D_\mu = \partial_\mu + ig_s T^a G^a_\mu + ig\tau^i W^i_\mu + ig' Y B_\mu \tag{145}$$

21.2 Gauge Group

$$\boxed{G_{\text{SM}} = SU(3)_c \times SU(2)_L \times U(1)_Y} \tag{146}$$

This is the **Standard Model gauge group** with three factors:

1. $SU(3)_c$ = color symmetry (strong interactions, QCD)
2. $SU(2)_L$ = weak isospin (left-handed weak interactions)
3. $U(1)_Y$ = hypercharge (electromagnetic and weak)

Group	Field	Coupling	Index Range
$SU(3)_c$	G^a_μ (gluons)	g_s	$a = 1, \ldots, 8$
$SU(2)_L$	W^i_μ (weak bosons)	g	$i = 1, 2, 3$
$U(1)_Y$	B_μ (hypercharge)	g'	(single field)

Table 1: Standard Model gauge fields and couplings

21.3 Gauge Fields and Couplings

21.4 Field Strength Tensors

21.4.1 SU(3) Gluon Field Strength

$$G^a_{\mu\nu} = \partial_\mu G^a_\nu - \partial_\nu G^a_\mu + g_s f^{abc} G^b_\mu G^c_\nu \tag{147}$$

where f^{abc} are $SU(3)$ structure constants (see Section 19).

21.4.2 SU(2) Weak Field Strength

$$W^i_{\mu\nu} = \partial_\mu W^i_\nu - \partial_\nu W^i_\mu + g\epsilon^{ijk} W^j_\mu W^k_\nu \tag{148}$$

where ϵ^{ijk} is the Levi-Civita symbol ($\epsilon^{123} = 1$).

21.4.3 U(1) Hypercharge Field Strength

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{149}$$

This is Abelian (no self-interaction term).

21.5 Generators

21.5.1 SU(3) Generators

$$T^a = \frac{\lambda^a}{2}, \quad a = 1, \ldots, 8 \tag{150}$$

where λ^a are Gell-Mann matrices (see Section 19).
Normalization:

$$\mathrm{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab} \tag{151}$$

21.5.2 SU(2) Generators

$$\tau^i = \frac{\sigma^i}{2}, \quad i = 1, 2, 3 \tag{152}$$

where σ^i are Pauli matrices:

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{153}$$

Commutation relations:

$$[\tau^i, \tau^j] = i\epsilon^{ijk}\tau^k \tag{154}$$

Normalization:

$$\mathrm{Tr}(\tau^i \tau^j) = \frac{1}{2}\delta^{ij} \tag{155}$$

21.5.3 U(1) Generator

$$Y = \text{hypercharge operator} \tag{156}$$

For fermions:

$$Y = Q - T_3 \tag{157}$$

where Q is electric charge and $T_3 = \tau^3$ is the third component of weak isospin.

21.6 Electroweak Unification

21.6.1 Electroweak Lagrangian

Combining $SU(2)_L \times U(1)_Y$:

$$\mathcal{L}_{\text{EW}} = \text{Tr}\left[(D_\mu\Theta)^\dagger(D^\mu\Theta)\right] - \frac{1}{4}W^i_{\mu\nu}W^{i\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} \tag{158}$$

21.6.2 Physical Gauge Bosons

After electroweak symmetry breaking, the physical bosons are:

$$W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp iW^2_\mu) \tag{159}$$

$$Z^0_\mu = \cos\theta_W W^3_\mu - \sin\theta_W B_\mu \tag{160}$$

$$A_\mu = \sin\theta_W W^3_\mu + \cos\theta_W B_\mu \tag{161}$$

where θ_W is the **weak mixing angle** (Weinberg angle).

21.6.3 Weak Mixing Angle

$$\tan\theta_W = \frac{g'}{g} \tag{162}$$

Experimental value:

$$\sin^2\theta_W \approx 0.23122 \tag{163}$$

In UBT, this is **derived** from the internal structure of Θ (see Section **??**).

21.7 Coupling Constants

21.7.1 Gauge Couplings

Coupling	**Symbol**	**Approximate Value**
Strong coupling	$\alpha_s(M_Z)$	≈ 0.118
Weak coupling	$\alpha_2 = g^2/(4\pi)$	$\approx 1/30$
Hypercharge coupling	$\alpha_1 = g'^2/(4\pi)$	$\approx 1/59$
Electromagnetic	$\alpha = e^2/(4\pi)$	$\approx 1/137.036$

Table 2: Standard Model coupling constants at M_Z

21.7.2 Electromagnetic Coupling

The electromagnetic coupling relates to weak couplings via:

$$\frac{1}{\alpha} = \frac{1}{\alpha_2} + \frac{1}{\alpha_1} \tag{164}$$

or equivalently:

$$e = g \sin\theta_W = g' \cos\theta_W \tag{165}$$

21.8 Fermion Representations

21.8.1 Quark Doublets (Left-Handed)

$$Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \quad Q'_L = \begin{pmatrix} c_L \\ s_L \end{pmatrix}, \quad Q''_L = \begin{pmatrix} t_L \\ b_L \end{pmatrix} \tag{166}$$

Quantum numbers:

$$(SU(3), SU(2), Y) = (\mathbf{3}, \mathbf{2}, +\frac{1}{6}) \tag{167}$$

21.8.2 Quark Singlets (Right-Handed)

$$u_R, c_R, t_R : \quad (\mathbf{3}, \mathbf{1}, +\frac{2}{3}) \tag{168}$$

$$d_R, s_R, b_R : \quad (\mathbf{3}, \mathbf{1}, -\frac{1}{3}) \tag{169}$$

21.8.3 Lepton Doublets (Left-Handed)

$$L_L = \begin{pmatrix} \nu_e \\ e_L \end{pmatrix}, \quad L'_L = \begin{pmatrix} \nu_\mu \\ \mu_L \end{pmatrix}, \quad L''_L = \begin{pmatrix} \nu_\tau \\ \tau_L \end{pmatrix} \tag{170}$$

Quantum numbers:

$$(SU(3), SU(2), Y) = (\mathbf{1}, \mathbf{2}, -\frac{1}{2}) \tag{171}$$

21.8.4 Lepton Singlets (Right-Handed)

$$e_R, \mu_R, \tau_R : \quad (\mathbf{1}, \mathbf{1}, -1) \tag{172}$$

21.9 Higgs Mechanism

21.9.1 Higgs Doublet

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \quad (SU(3), SU(2), Y) = (\mathbf{1}, \mathbf{2}, +\frac{1}{2}) \tag{173}$$

21.9.2 Higgs Potential

$$V(\Phi) = -\mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2 \tag{174}$$

21.9.3 Vacuum Expectation Value

$$\langle\Phi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}, \quad v \approx 246 \text{ GeV} \tag{175}$$

21.9.4 Gauge Boson Masses

$$M_W = \frac{gv}{2} \approx 80.4 \text{ GeV} \tag{176}$$

$$M_Z = \frac{v}{2}\sqrt{g^2 + g'^2} = \frac{M_W}{\cos\theta_W} \approx 91.2 \text{ GeV} \tag{177}$$

$$M_\gamma = 0 \quad \text{(photon remains massless)} \tag{178}$$

21.10 Yukawa Couplings

Fermion masses arise from Yukawa interactions:

$$\mathcal{L}_{\text{Yukawa}} = -y_u \bar{Q}_L \tilde{\Phi} u_R - y_d \bar{Q}_L \Phi d_R - y_e \bar{L}_L \Phi e_R + \text{h.c.} \tag{179}$$

where $\tilde{\Phi} = i\sigma^2\Phi^*$ and y_u, y_d, y_e are Yukawa coupling matrices.

Fermion masses:

$$m_f = \frac{y_f v}{\sqrt{2}} \tag{180}$$

21.11 CKM Matrix

Quark mixing is described by the Cabibbo-Kobayashi-Maskawa (CKM) matrix:

$$V_{\text{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \tag{181}$$

Unitarity triangle relations provide CP violation tests.

21.12 PMNS Matrix

Neutrino mixing is described by the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix:

$$U_{\text{PMNS}} = \begin{pmatrix} U_{e1} & U_{e2} & U_{e3} \\ U_{\mu 1} & U_{\mu 2} & U_{\mu 3} \\ U_{\tau 1} & U_{\tau 2} & U_{\tau 3} \end{pmatrix} \tag{182}$$

21.13 Emergence in UBT

21.13.1 Gauge Group from Theta Field (Candidate Construction)

In UBT, the SM gauge group is **conjectured** to arise from the 8×8 extended Θ field as a candidate geometric mechanism (full derivation pending; see `docs/sm_embedding_roadmap.md`):

$$\Theta_{8\times 8} \quad \xrightarrow{\text{conjecture}} \quad SU(3)_c \times SU(2)_L \times U(1)_Y \tag{183}$$

The decomposition:

$$\Theta = \sum_a \Theta_a^{\text{color}} \otimes T^a + \sum_i \Theta_i^{\text{weak}} \otimes \tau^i + \Theta^Y \otimes Y \tag{184}$$

naturally produces the SM gauge structure.

21.13.2 Predictions

From the Θ field geometry, UBT **aims to derive** (currently open problems — not yet accomplished):

- $\sin^2\theta_W$ — [**Semi-empirical, not derived from** $S[\Theta]$]
- $\alpha_s(M_Z)$ — [**Open**]
- Fermion mass ratios — [**Open Hard Problem**]
- CKM/PMNS matrix elements — [**Open**]

21.14 Grand Unification

21.14.1 GUT Scale

At high energies ($E \sim 10^{16}$ GeV), the three couplings may unify:

$$\alpha_s(M_{\text{GUT}}) = \alpha_2(M_{\text{GUT}}) = \alpha_1(M_{\text{GUT}}) = \alpha_{\text{GUT}} \tag{185}$$

Possible GUT groups:

- $SU(5)$
- $SO(10)$
- E_6

21.14.2 UBT and GUT

The Θ field structure may naturally accommodate GUT symmetries at high energies while breaking to $SU(3) \times SU(2) \times U(1)$ at low energies.

21.15 Conflict Resolution

This canonical definition supersedes all previous inconsistent SM formulations by:

- Unifying notation for all three gauge groups
- Standardizing generator normalizations
- Consistent index conventions throughout
- Embedding in Θ field structure

21.16 Experimental Status

All SM predictions have been confirmed by experiment, including:

- Higgs boson discovery (2012, $m_H \approx 125$ GeV)
- Precision electroweak tests
- Quark and lepton masses
- CKM matrix elements
- Neutrino oscillations (PMNS matrix)

UBT must reproduce all these results in the limit $\psi \to 0$.

21.17 Historical Note

This canonical SM gauge structure provides the complete Standard Model within UBT, where all gauge symmetries emerge from the fundamental Θ field rather than being postulated *a priori*.

21.18 Geometric Derivation of $SU(2)_L \times U(1)_Y$

The emergence of $SU(2)_L \times U(1)_Y$ from the $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{H}$ algebra is derived explicitly in Appendix E2 (`consolidation_project/appendix_E2_SM_geometry.tex`, Section **??**).

Key results:

- $SU(2)_L$ arises as the norm-preserving, unit-determinant *left action* on $\mathrm{Mat}(2,\mathbb{C})$: generators $T^a : M \mapsto \frac{i\sigma^a}{2} M$. Commutation relations $[T^a, T^b] = \varepsilon^{abc} T^c$ are verified. Status: [**DERIVED**].
- $U(1)_Y$ arises as the scalar phase *right action*: $\Theta \mapsto e^{-i\theta}\Theta$. Status: [**DERIVED**].
- The Weinberg angle θ_W *cannot* be fixed by $\mathbb{C} \otimes \mathbb{H}$ geometry alone; it remains a [**SEMI-EMPIRICAL**] free parameter.
- Chirality ($SU(2)_L$ only) is motivated by ψ-parity selection (Option A) but not yet derived; status: [**SEMI-EMPIRICAL**].

21.19 Higgs Mechanism via Hosotani Wilson Line

The electroweak Higgs mechanism arises in UBT from the Hosotani mechanism on the ψ-circle.

Theorem (Hosotani SSB condition, [L1]). At KK level $n_R = 138$ (even winding), all 8 degrees of freedom of $\mathbb{C} \otimes \mathbb{H}$ are bosonic (KK parity rule: even $n \to$ bosonic; see `fermionic_statistics_bridge.tex`). Therefore $N_{\text{eff}} = N_{\text{bos}} - N_{\text{fer}} = 8 > 0$. The Wilson line effective potential $V(\theta_W) \propto -N_{\text{eff}} \cos(2\pi\theta_W)$ has minimum at $\theta_W = \pi/2$ in this effective description. The earlier $p = 139$ mirror-sector interpretation has been moved to the speculative extension and is not used here as a canonical input. Spontaneous symmetry breaking occurs: $\langle\theta_W\rangle = \pi/2 \neq 0$.

Higgs VEV estimate (v67):

$$v \approx \frac{n_R\,\pi}{2\,g\,R_\psi} \approx 230\ \text{GeV} \qquad (n_R = 138,\ g \approx 0.65,\ R_\psi \approx 14.18\ \text{GeV}^{-1}). \tag{186}$$

Quartic coupling estimate (v67): After full 5D→4D reduction (hosotani_higgs.tex §7),

$$\lambda_{4\text{D}} = \frac{3\,N_{\text{eff}}\,\zeta(3)}{4\pi R_\psi^2} \approx 0.0114 \qquad (\text{SM: } \lambda_{\text{SM}} \approx 0.13;\ \text{factor} \sim 11). \tag{187}$$

Status: [**L1**] — N_{eff} count and SSB condition proved. $\lambda_{4\text{D}}$ estimate: [**L1**] (correct order of magnitude, factor ~ 11 from λ_{SM}; remaining corrections identified in `research_tracks/research/hoso`
Numerical verification: `tools/compute_hosotani_lambda.py`.

22 Theta-Functions and Toroidal Structure

22.1 Jacobi Theta Functions

22.2 Modular Transformations

22.3 Fine Structure Constant

The fine structure constant emerges from the toroidal geometry:

$$\alpha \approx \frac{1}{137.036} \tag{188}$$

23 Speculative Extensions

Note: Speculative Content Quarantined

The following speculative interpretations (psychons, consciousness coupling, Theta-resonator) are **not part of the core physics** and have been moved to `speculative_extensions/consciousness/` (psychon content) and `speculative_extensions/complex_consciousness/` (broader consciousness models).
The imaginary sector of UBT (ψ, χ, ξ components) is a well-defined **internal auxiliary sector** from the perspective of core physics. Any speculative physical interpretation of these components is strictly separated here.

24 Experimental Designs and Testable Predictions

24.1 Testable Predictions

UBT makes specific, falsifiable predictions:

1. Fine structure constant: $\alpha = 1/137.035999...$
2. Electron mass: $m_e \approx 0.511$ MeV

3. Muon/tau mass ratios
4. QCD scale: $\Lambda_{\text{QCD}} \approx 200$ MeV
5. Neutrino masses
6. Dark matter signatures
7. Consciousness-correlated phenomena

24.2 Experimental Status

25 Conclusions

The Unified Biquaternion Theory provides a geometric framework unifying General Relativity, Quantum Field Theory, and the Standard Model. Key achievements:

- **Unification**: Single field $\Theta(q, \tau)$ generates geometry and matter
- **GR compatibility**: Exact reproduction of Einstein's equations
- **SM emergence**: Gauge groups emerge from Θ structure
- **Predictions**: Fundamental constants derived, not postulated
- **Testability**: Specific experimental predictions

Future work includes:

1. Detailed cosmological implications
2. Quantum gravity regime calculations
3. Experimental validation programs
4. Extension to dark sector physics

Acknowledgments

A Structure of the Covariant Derivative ∇

B The covariant derivative in the active UBT geometry

The compact master-equation notation

$$D^\dagger D\Theta = \kappa\mathcal{T}$$

is meaningful only after the representation and multiplication sides of D are specified. A covariant derivative compares field values expressed in local bases that may rotate or boost from point to point. An ordinary partial derivative sees both the physical change of the field and the change of the basis; the connection term compensates the latter.

B.1 Three distinct connections

The symbols must not be conflated:

- ${\Gamma^\rho}_{\mu\nu}$ transports coordinate/spacetime indices;
- ${\omega_\mu}^a{}_b$ transports local Lorentz-frame indices;
- $\Omega_\mu = \rho_*(\omega_\mu)$ is the spin/biquaternionic representation acting on the chosen field space.

They are linked by the tetrad-compatibility equation

$$\partial_\mu {e_\nu}^a - {\Gamma^\rho}_{\mu\nu} {e_\rho}^a + {\omega_\mu}^a{}_b {e_\nu}^b = 0,$$

not by taking a componentwise real part.

For specified tetrad and torsion,

$$\omega = \mathring{\omega}(e) + K(T), \qquad K_{abc} = \tfrac{1}{2}(T_{cab} - T_{abc} - T_{bca}).$$

Thus the torsion-free classical branch has the Levi–Civita spin connection. The canonical action must still determine whether torsion vanishes or is sourced by spin/internal structure.

B.2 Why the generic curved derivative is two-sided

Biquaternions act naturally from both sides. The active curved candidate is

$$\boxed{D_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta - \Theta B_\mu.}$$

Its curvature identity is

$$\boxed{[D_\mu, D_\nu]\Theta = F^A_{\mu\nu}\Theta - \Theta F^B_{\mu\nu}.}$$

A one-sided derivative $D^L_\mu\Theta = \partial_\mu\Theta + A_\mu\Theta$ is allowed in flat or special sectors, but under invertibility and torsion-free antisymmetric tetrad compatibility it forces its curvature to vanish. It therefore cannot be assumed as the generic curved-GR representation.

The precise decomposition of (A_μ, B_μ) into Lorentz-spin and Standard Model gauge actions remains an action-level problem. It must be derived, not inserted as the assertion that all interactions automatically live in one operator.

B.3 Special relativity and the implicit curved system

In Cartesian inertial Minkowski gauge,

$$\Gamma = \omega = \Omega = 0, \qquad D_\mu = \partial_\mu,$$

and

$$\Theta_{\mathrm{SR}} = \Theta_0 + \sqrt{\mathcal{N}_0}(ix^0\mathbf{1} + x^k\mathbf{e}_k)$$

generates the Minkowski tetrad explicitly.

In curved geometry the reconstructed connections depend on the tetrad that is itself generated by $D\Theta$. Hence

$$E_\mu = \mathcal{N}_0^{-1/2}[\partial_\mu\Theta + A_\mu[E,T]\Theta - \Theta B_\mu[E,T]]$$

is an implicit nonlinear first-order PDE/fixed-point system. A Jacobi-theta restriction may make the complete system transcendental as well; that does not remove the separate existence, uniqueness, regularity, and integrability questions.

C Symbol Dictionary

D Symbol Dictionary and Notation Conventions

D.1 Purpose

This section establishes the **unique, canonical meaning** of all symbols used in the Unified Biquaternion Theory. Each symbol has **exactly one meaning** to avoid confusion and ensure consistency across all documents.

D.2 Reserved Symbols — Single Meaning Only

Symbol	Canonical Meaning	Notes
α	Fine structure constant $\approx 1/137.036$	NO other uses (angle, decay rate, etc.)
ψ	Imaginary component of complex time	NOT wavefunction, NOT spinor
τ	Complex time $= t + i\psi$	NOT proper time
Θ	Fundamental biquaternion field	Capital theta only for field
E_μ	Covariant UBT tetrad	$E_\mu = \mathcal{N}_0^{-1/2} D_\mu \Theta$
${\Gamma^\rho}_{\mu\nu}$	Coordinate/affine connection	Acts on spacetime indices
${\omega_\mu}^a{}_b$	Lorentz-frame connection	$\omega = \mathring{\omega}(e) + K(T)$ for specified torsion
Ω_μ	Spin/biquaternionic lift of ω_μ	Representation-dependent; not $\mathrm{Re}^{-1}\,\Gamma$
q	Biquaternion coordinate (4 DOF)	NOT charge
$\mathcal{G}_{\mu\nu}$	Historical/candidate biquaternionic bilinear	Does not define the active metric
$g_{\mu\nu}$	Lorentz metric tensor	Central coefficient of $\{E_\mu, E_\nu\}_\sharp/2 = g_{\mu\nu}\mathbf{1}$
$\mathcal{T}_{\mu\nu}$	Biquaternionic stress-energy tensor	Fundamental (geometric phase response)
$T_{\mu\nu}$	Classical stress-energy tensor	DERIVED: $T_{\mu\nu} := \mathrm{Re}(\mathcal{T}_{\mu\nu})$
$F_{\mu\nu}$	Electromagnetic field strength	QED only
$G^a_{\mu\nu}$	Gluon field strength	QCD only
$W^i_{\mu\nu}$	Weak field strength	Weak interactions
$B_{\mu\nu}$	Hypercharge field strength	Electroweak

Table 3: Reserved symbols with unique canonical meanings

D.3 Index Conventions

D.3.1 Spacetime Indices

- **Greek indices** $\mu, \nu, \rho, \sigma, \lambda = 0, 1, 2, 3$ for spacetime coordinates
- **Coordinates**: $x^\mu = (x^0, x^1, x^2, x^3) = (t, x, y, z)$ or (ct, x, y, z)

- **Summation convention**: Repeated indices are summed (Einstein convention)

D.3.2 Spatial Indices

- **Latin indices** $i, j, k = 1, 2, 3$ for spatial coordinates only
- **Coordinates**: $x^i = (x^1, x^2, x^3) = (x, y, z)$

D.3.3 Gauge Indices

- **Color indices** $a, b, c = 1, 2, \ldots, 8$ for $SU(3)$ adjoint representation
- **Weak isospin indices** $i, j, k = 1, 2, 3$ for $SU(2)$ generators
- **Flavor indices** $f, g = u, d, s, c, b, t$ for quark flavors
- **Generation indices** $\alpha, \beta = 1, 2, 3$ for fermion generations

D.3.4 Matrix Indices

- **Matrix elements** $A, B, C = 0, 1, 2, 3$ for Θ components: Θ_{AB}
- **Internal indices** $m, n = 1, 2, \ldots, N$ for general matrix dimensions

D.4 Field and Operator Notation

Symbol	Meaning	Context
$\Theta(q, \tau)$	Fundamental biquaternion field	Core UBT field
$\Theta^\dagger$	Hermitian conjugate of Θ	$(\bar{\Theta})^T$
Ψ	Wavefunction (if needed)	Use capital psi
A_μ	Electromagnetic potential	QED photon field
G^a_μ	Gluon field	QCD, $a = 1, \ldots, 8$
W^i_μ	Weak boson field	Electroweak, $i = 1, 2, 3$
B_μ	Hypercharge field	Electroweak
Φ	Higgs field	Electroweak symmetry breaking

Table 4: Field and operator notation

D.5 Derivative Notation

Symbol	Meaning	Notes
∂_μ	Partial derivative $\frac{\partial}{\partial x^\mu}$	Coordinate derivative
∂^μ	Contravariant derivative $g^{\mu\nu}\partial_\nu$	Raised index
∇_μ	Covariant derivative (gravity)	Includes Christoffel symbols
D_μ	Gauge covariant derivative	Includes gauge connection
$\nabla^\dagger$	Biquaternionic conjugate derivative	UBT-specific

Symbol	Meaning	Notes
$\Box$	d'Alembertian $\partial^\mu \partial_\mu$	Wave operator

Table 5: Derivative notation

D.6 Coupling Constants

Symbol	Meaning	Value	Status
α	Fine structure constant	$\approx 1/137.036$	Predicted
α_s	Strong coupling	≈ 0.118 at M_Z	Predicted
g	Weak coupling	$SU(2)_L$	Derived
g'	Hypercharge coupling	$U(1)_Y$	Derived
g_s	Strong coupling	$SU(3)_c$	Derived
e	Elementary charge	$\sqrt{4\pi\alpha}$	Input
G	Newton's constant	6.674×10^{-11} $\mathrm{m^3/kg/s^2}$	Input

Table 6: Coupling constants

D.7 Mass Scales

Symbol	Meaning	Approximate Value	Status
m_e	Electron mass	0.511 MeV	Predicted
m_μ	Muon mass	105.7 MeV	Predicted
m_τ	Tau mass	1.777 GeV	Predicted
m_p	Proton mass	938.3 MeV	Input
Λ_{QCD}	QCD scale	~ 200 MeV	Predicted
M_W	W boson mass	80.4 GeV	Derived
M_Z	Z boson mass	91.2 GeV	Derived
m_H	Higgs mass	125 GeV	Input
M_{Pl}	Planck mass	1.22×10^{19} GeV	Fundamental

Table 7: Mass scales

D.8 Forbidden Symbol Uses

To maintain clarity and avoid conflicts, the following uses are **explicitly forbidden**:

1. α for any angle, decay rate, or parameter other than fine structure constant
2. ψ for wavefunction (use Ψ instead) or spinor (use ψ_{spinor})
3. τ for proper time (use s or λ)
4. q for electric charge (use Q or e)
5. Θ (lowercase theta) for field (reserved for angles if needed)

6. Using $g_{\mu\nu}$ as fundamental (must derive from $\mathcal{G}_{\mu\nu}$)

7. Using $T_{\mu\nu}$ as fundamental (must derive from $\mathcal{T}_{\mu\nu}$)

8. Defining the metric by a trace, real-part, phase, fiber, or embedding readout instead of the central tetrad identity

D.9 Metric Signature Convention

Default signature: $(+,-,-,-)$ (mostly minus, timelike positive)

- $g_{00} > 0$ (timelike positive)
- $g_{11}, g_{22}, g_{33} < 0$ (spacelike negative)

Alternative signature $(-,+,+,+)$ may be used in specific contexts with explicit notice.

D.10 Unit Conventions

D.10.1 Natural Units

Default: $\hbar = c = 1$

- Energy, mass, temperature have dimension [mass]
- Length, time have dimension $[\text{mass}]^{-1}$
- Action is dimensionless

D.10.2 SI Units

When presenting results:

- Masses in eV, keV, MeV, GeV
- Lengths in meters, nm, fm
- Times in seconds
- Energies in eV, joules

D.11 Special Function Notation

Symbol	**Meaning**	**Notes**
$\mathrm{Tr}(A)$	Matrix trace	$\sum_i A_{ii}$
$\det(A)$	Matrix determinant	Determinant
$\mathrm{Re}(z)$	Real part	Real component
$\mathrm{Im}(z)$	Imaginary part	Imaginary component
$\bar{z}$	Complex conjugate	Conjugation
$\theta_i(z,\tau)$	Jacobi theta functions	$i = 1,2,3,4$
$\zeta(s)$	Riemann zeta function	Number theory

Symbol	Meaning	Notes

Table 8: Special function notation

D.12 Tensor and Matrix Operations

Operation	Notation	Meaning
Matrix product	AB	$(AB)_{ij} = \sum_k A_{ik}B_{kj}$
Tensor product	$A \otimes B$	Outer product
Commutator	$[A, B]$	$AB - BA$
Anticommutator	$\{A, B\}$	$AB + BA$
Covariant derivative	$\nabla_\mu T_{\nu\rho}$	Includes connection
Lie derivative	$\mathcal{L}_X T$	Along vector field X

Table 9: Tensor and matrix operations

D.13 Abbreviations and Acronyms

Acronym	Meaning
UBT	Unified Biquaternion Theory
GR	General Relativity
QFT	Quantum Field Theory
QED	Quantum Electrodynamics
QCD	Quantum Chromodynamics
SM	Standard Model
EW	Electroweak
GUT	Grand Unified Theory
CKM	Cabibbo-Kobayashi-Maskawa (quark mixing matrix)
PMNS	Pontecorvo-Maki-Nakagawa-Sakata (neutrino mixing matrix)
EWSB	Electroweak Symmetry Breaking
VEV	Vacuum Expectation Value
DOF	Degrees of Freedom

Table 10: Abbreviations and acronyms

D.14 Version Control

This symbol dictionary is version-controlled and canonical. Any proposed changes must:

1. Be justified by theoretical necessity
2. Not conflict with existing usage
3. Be documented in changelog
4. Update all affected documents

D.15 Enforcement

All UBT documents must:

- Comply with this symbol dictionary
- Report conflicts as errors
- Reference this section for definitions
- Avoid introducing new symbol meanings

E Mathematical Derivations

F Computational Methods